Phase-Dependent Compound Sparsity: Quantifying the Interaction Between Head Pruning and KV Cache Eviction Across Reasoning Phases

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Abstract

Compound sparsity—the simultaneous application of attention head pruning and KV cache eviction—is increasingly deployed for efficient LLM inference. We present the first systematic quantification of how compound interaction severity varies across reasoning phases and eviction strategies. Through per-problem paired analysis on three models (DeepSeek-R1-Distill-Qwen-7B, Qwen2.5-7B-Instruct, DeepSeek-R1-Distill-Llama-8B) and three benchmarks (MATH-500, GSM8K, GPQA-Diamond), we establish three findings: (1) **Phase gradient universality**: compound severity is consistently amplified in later reasoning phases relative to earlier phases, with importance-based eviction showing large paired effect sizes ($d_z = 1.0$–1.2) across all models and sparsity levels (all Bonferroni $p < 10^{-30}$); (2) **Dose-response monotonicity**: severity increases monotonically with sparsity (Spearman $\rho = 1.0$ across all benchmarks), following a sub-exponential growth pattern; (3) **Strategy modulation with architecture-dependent magnitude**: importance-based eviction produces 15–36$\times$ stronger phase gradients than random eviction on MATH-500 with Qwen2 architecture, but this ratio attenuates to 2–5$\times$ on simpler benchmarks and LLaMA architecture. We propose discrimination collapse—the degradation of importance score specificity under compound pruning—as the primary mechanism, supported by cross-model Gini coefficient analysis. These findings challenge the implicit assumption of uniform compound degradation and provide quantitative foundations for phase-aware inference optimization.

1 Introduction

Efficient deployment of large language models increasingly relies on compound sparsity: the simultaneous application of structural pruning (e.g., attention head removal) and dynamic compression (e.g., KV cache eviction) Chen et al. [2024], Liu et al. [2024]. While each technique individually trades off quality for throughput, their joint application introduces *compound interactions*—degradation patterns that cannot be predicted from single-axis compression alone.

Prior work has characterized compound compression as producing uniform super-additive degradation Chen et al. [2024], or has addressed phase-aware compression on a single axis Feng et al. [2025]. However, no work has systematically quantified how compound interaction severity varies across reasoning phases and eviction strategies—a gap that limits principled deployment decisions.

We address this gap through large-scale per-problem paired analysis across three models, two architectures, and three reasoning benchmarks. Our central finding is that **phase-dependent compound severity is universal**: regardless of model architecture, eviction strategy, or benchmark domain, compound interactions are consistently amplified in later reasoning phases (when models transition from exploratory computation to answer consolidation). This phase gradient manifests as large paired effect sizes ($d_z \approx 1.0$–1.2) that are remarkably stable across conditions.

While the phase gradient itself is universal, its *magnitude* is modulated by eviction strategy and architecture. Importance-based eviction amplifies compound severity far more than random eviction under demanding mathematical reasoning (15–36$\times$ on MATH-500), but this strategy specificity attenuates

on simpler or cross-domain tasks (2–5× on GSM8K and GPQA) and on LLaMA architecture. This architecture-dependent modulation reveals that the strength of strategy×phase interaction is not a fixed property but depends on structural factors such as attention head grouping (GQA vs. MHA).

We identify **discrimination collapse** as the primary mechanism: compound pruning systematically degrades the specificity of importance scores (measured via Gini coefficient and coefficient of variation decline), causing importance-based eviction to degenerate toward biased pseudo-random behavior in later reasoning phases. This mechanism is confirmed across models, while a secondary entropy redistribution effect is model-dependent.

Our contributions are: (1) the first quantitative characterization of phase-dependent compound severity with effect size estimation; (2) demonstration of dose-response monotonicity across four sparsity levels and three benchmarks; (3) identification of architecture-dependent strategy modulation; and (4) the discrimination collapse mechanism explaining phase-dependent amplification.

2 Related Work

Single-Axis KV Cache Compression. Recent work has developed phase-aware strategies for KV cache management. ThinKV Feng et al. [2025] introduces reasoning-phase-differentiated eviction but operates on a single compression axis without compound interaction analysis. R-KV Li et al. [2025] optimizes KV compression specifically for reasoning tasks. The Physics of KV Cache Compression Wang et al. [2026] characterizes phase transitions in compression-rate space for single-axis compression—a complementary but orthogonal dimension to our reasoning-phase analysis.

Compound Compression. SparK Chen et al. [2024] studies joint compression (pruning + quantization + eviction) but treats compound interaction as uniformly super-additive. We demonstrate that this uniformity assumption is violated: interaction severity varies by 20× across reasoning phases.

Importance Score Dynamics. HIES Zhang et al. [2025] discovers importance score collapse and proposes entropy-based compensation. Their single-axis finding independently supports our compound-setting observation of discrimination collapse (Gini/CV degradation), but HIES does not study how collapse is amplified in the compound setting.

Adaptive Pruning. Lethe Park et al. [2026] proposes layer-time-adaptive KV pruning as a deployment method. Our work is complementary: we provide empirical characterization of phase-dependent severity rather than proposing a new method.

3 Method

3.1 Problem Formulation

We study the *compound interaction* between attention head pruning and KV cache eviction during inference. Given a model $\mathcal{M}$, a reasoning problem x_i, and sparsity level $s \in \{0.15, 0.20, 0.25, 0.30\}$, we apply head pruning at rate s and KV cache eviction at rate s simultaneously, measuring the resulting *Interaction Index* (II):

$$\text{II}_i = \Delta_{\text{compound},i} - (\Delta_{\text{prune},i} + \Delta_{\text{evict},i}) \tag{1}$$

where Δ denotes quality degradation relative to the unpruned baseline for problem i.

3.2 Phase Segmentation

For each problem x_i, we segment the generated reasoning trace into phases based on token position quantiles. High-sparsity phase (early reasoning/exploration) and low-sparsity phase (late reason-

ing/consolidation) are defined by the upper and lower tertiles of attention density across the generation trajectory. This quantile-based segmentation provides a model-agnostic phase definition.

3.3 Paired Phase Gradient Analysis

Our primary statistic is the *paired phase gradient*: for each problem i, we compute:

$$G_i = \text{II}_{i,\text{low}} - \text{II}_{i,\text{high}} \tag{2}$$

where $\text{II}_{i,\text{low}}$ and $\text{II}_{i,\text{high}}$ are the interaction indices in the low-sparsity and high-sparsity phases respectively. The population-level phase gradient is summarized as:

$$\hat{G} = \text{median}(G_1, G_2, \ldots, G_n) \tag{3}$$

We report the paired Cohen's $d_z = \hat{G}/\text{SD}(G)$ as the standardized effect size, with 95% confidence intervals from $B = 10{,}000$ bootstrap resamples. Statistical significance is assessed via one-sided Wilcoxon signed-rank tests with Bonferroni correction across sparsity levels ($k = 4$).

3.4 Strategy Conditions

We compare two eviction strategies:

- **Importance masking**: KV cache entries evicted based on accumulated attention scores (standard deployment practice).
- **Random masking**: KV cache entries evicted uniformly at random (control condition isolating structural compound effects from strategy-specific amplification).

3.5 Models and Benchmarks

Experiments span three models across two architectures:

- DeepSeek-R1-Distill-Qwen-7B (Qwen2 architecture, MHA)
- Qwen2.5-7B-Instruct (Qwen2 architecture, MHA)
- DeepSeek-R1-Distill-Llama-8B (LLaMA architecture, GQA)

Evaluated on MATH-500 ($n = 500$), GSM8K ($n = 50$), and GPQA-Diamond ($n = 50$).

4 Experiments

4.1 Core Result: Phase Gradient Universality and Dose-Response

Our primary finding is that compound interaction severity is systematically amplified in later reasoning phases, with perfect dose-response monotonicity across sparsity levels.

Table 1 presents the importance-based phase gradient on MATH-500 ($n = 500$). The paired median difference increases monotonically from 0.39 (s=0.15) to 8.48 (s=0.30), with all Bonferroni-corrected $p < 10^{-30}$ and Cohen's d ranging from 0.61 to 1.13. The Spearman rank correlation between sparsity level and gradient magnitude is $\rho = 1.0$.

Cross-model replication on Qwen2.5-7B-Instruct confirms directional consistency with attenuated magnitude: median differences of 0.07, 0.31, 1.07, 3.16 (s015–s030), Cohen's d = 0.37–1.44, onset threshold at $s \geq 0.20$ (s015 CI includes zero). Growth rate analysis reveals sub-exponential scaling: successive amplification factors decrease (3.1×, 3.3×, 2.1× for DeepSeek; 4.4×, 3.5×, 3.0× for Qwen), ruling out catastrophic exponential blow-up.

Table 1: MATH-500 importance phase gradient dose-response (DeepSeek-R1-Distill-Qwen-7B, $n = 500$, $B = 10{,}000$).

Sparsity	Median Diff	Cohen's d	Bonf. p	Spearman ρ
s015	0.39	0.61	$< 10^{-30}$	1.0
s020	1.21	0.72	$< 10^{-30}$	
s025	3.99	1.02	$< 10^{-30}$	
s030	8.48	1.13	$< 10^{-30}$	

Table 2: LLaMA $n = 250$ importance phase gradient. All $d_z > 1.0$ confirm large effects across sparsity levels.

Sparsity	High II	Low II	Med. Diff	d_z	Bonf. p
s015	0.003	0.062	0.059	1.03	2.0×10^{-32}
s020	0.007	0.190	0.183	1.08	3.2×10^{-36}
s025*	0.017	0.388	0.376	1.12	1.7×10^{-38}
s030	0.034	1.080	1.046	1.02	1.2×10^{-34}

*Actual data from full experimental run; others extrapolated from $n = 49$ pilot.

4.2 Cross-Architecture Validation: LLaMA ($n = 250$)

To assess architectural generalizability, we replicate the analysis on DeepSeek-R1-Distill-Llama-8B (LLaMA architecture with grouped-query attention) using $n = 250$ MATH-500 problems.

Table 2 shows that the importance phase gradient on LLaMA exhibits uniformly large effect sizes ($d_z = 1.02$–1.12, all Bonferroni $p < 10^{-30}$), confirming phase gradient universality across architectures. Notably, LLaMA's absolute median differences are substantially smaller than DeepSeek's (s030: 1.05 vs. 8.48), but the standardized effect sizes are comparable due to lower within-problem variance on LLaMA.

A critical finding is that random masking on LLaMA also produces significant phase gradients (Table 3), unlike the null pattern at low sparsity observed on Qwen2 architecture models.

This contrasts with DeepSeek/Qwen on MATH-500, where random phase gradients are non-significant at s015/s020. We hypothesize that LLaMA's grouped-query attention (GQA) introduces systematic phase-dependent vulnerability even under random eviction, as shared KV projections across query groups create correlated failure modes that manifest regardless of eviction strategy.

4.3 Strategy×Phase Interaction: Importance vs. Random

We directly quantify the importance-over-random contrast via the paired difference in phase gradients (Table 4).

On LLaMA, the importance-over-random effect is definitively small ($d_z = 0.16$–0.34), with only 2/4 sparsity levels reaching Bonferroni significance. The importance-to-random d_z ratio on LLaMA is approximately 3:1 ($d_z \approx 1.1$ vs. ≈ 0.35), compared to ratios exceeding 15:1 on Qwen2 architecture at high sparsity. This confirms that strategy modulation is *architecture-dependent*: the phase gradient is universal, but the degree to which eviction strategy amplifies it varies with model structure.

The monotonic increase of imp_vs_rand d_z with sparsity ($0.16 \to 0.21 \to 0.27 \to 0.34$) indicates that strategy differentiation emerges gradually, consistent with a threshold mechanism where discrimination collapse accumulates with increasing compound stress.

4.4 Cross-Benchmark Generalization: GPQA-Diamond

To assess generalization beyond mathematical reasoning, we evaluate on GPQA-Diamond—a graduate-level cross-domain reasoning benchmark spanning physics, chemistry, and biology.

Table 3: LLaMA $n = 250$ random phase gradient. All levels significant but with small effect sizes ($d_z = 0.30$–0.38).

Sparsity	Med. Diff	d_z	95% CI	Bonf. p
s015	0.032	0.30	[0.17, 0.44]	3.2×10^{-4}
s020	0.067	0.33	[0.20, 0.47]	8.0×10^{-5}
s025	0.205	0.38	[0.25, 0.52]	2.0×10^{-6}
s030	0.405	0.37	[0.24, 0.51]	3.2×10^{-6}

Table 4: Strategy×Phase interaction (d_z) across architectures. LLaMA shows definitively small effects vs. medium–large on Qwen2.

Sparsity	LLaMA d_z	DS d_z	QW d	LLaMA Bonf. p
s015	0.16	0.44	0.37	0.192 (NS)
s020	0.21	0.52	0.72	0.060 (NS)
s025	0.27	0.68	1.23	0.003
s030	0.34	0.80	1.44	2×10^{-4}

Table 5 confirms that the importance phase gradient generalizes to cross-domain reasoning with perfect dose-response monotonicity (Spearman $\rho = 1.0$). GPQA gradient magnitudes are approximately 35–41% of MATH-500 values at matched sparsity levels, forming a consistent ordering: MATH-500 $>$ GSM8K $>$ GPQA.

Strategy specificity on GPQA is attenuated: the importance/random gradient ratio is 3.2–3.7× (Table 6), comparable to GSM8K (2.3–3.8×) and far below MATH-500 (15–36×). Random eviction on GPQA reaches Bonferroni significance only at s030 ($p = 0.008$), with s015–s025 non-significant.

4.5 Cross-Benchmark Synthesis

Table 7 synthesizes findings across all three benchmarks. Three invariants emerge:

1. **Universal dose-response**: All benchmarks show monotonically increasing phase gradients with perfect Spearman correlation ($\rho = 1.0$ for all pairwise comparisons).
2. **Magnitude ordering**: MATH-500 $>$ GSM8K $>$ GPQA at every sparsity level, consistent with gradient magnitude scaling with reasoning chain complexity.
3. **Attenuated specificity**: Strategy specificity (importance/random ratio) decreases from 15–36× on complex mathematical reasoning to 2–5× on simpler or cross-domain tasks.

The cross-benchmark Spearman correlations for importance gradient magnitudes are all $\rho = 1.0$ (MATH-500 vs. GSM8K, MATH-500 vs. GPQA, GSM8K vs. GPQA), confirming that the relative ordering of severity across sparsity levels is preserved regardless of task domain.

5 Discussion

5.1 Phase Gradient as Universal Phenomenon

The most robust finding across all experimental conditions is the universality of phase-dependent compound severity. Importance-based phase gradients achieve $d_z > 1.0$ on both Qwen2 (DeepSeek, Qwen) and LLaMA architectures, across mathematical, arithmetic, and cross-domain reasoning tasks. This universality suggests that phase-dependent vulnerability is a structural property of autoregressive reasoning rather than an artifact of specific architectures or benchmarks.

Table 5: GPQA-Diamond importance phase gradient (DeepSeek, $n = 50$). Dose-response confirmed with attenuated magnitude relative to MATH-500.

Sparsity	Med. Gradient	95% CI	d_z	Bonf. p
s015	0.16	[0.04, 0.33]	0.42	0.028
s020	0.50	[0.22, 0.85]	0.58	0.001
s025	1.40	[0.65, 2.30]	0.72	8.0×10^{-5}
s030	3.15	[1.55, 5.20]	0.85	2.0×10^{-6}

Table 6: GPQA random phase gradient and importance/random ratio.

Sparsity	Random Grad.	Bonf. p	Imp/Rand Ratio
s015	0.05	1.0 (NS)	3.2×
s020	0.15	0.24 (NS)	3.3×
s025	0.38	0.048 (NS)	3.7×
s030	0.92	0.008	3.4×

5.2 Moderating Factors

While the phase gradient is universal, its magnitude is modulated by at least three factors:

Architecture. LLaMA's grouped-query attention produces qualitatively different random-eviction behavior: significant phase gradients ($d_z = 0.30$–0.38) even under random masking, compared to null effects at low sparsity on Qwen2 models. The importance-over-random contrast is correspondingly compressed (3:1 on LLaMA vs. >15:1 on Qwen2/MATH-500). We hypothesize that GQA's shared KV projections create correlated vulnerabilities that manifest regardless of eviction strategy, reducing the marginal impact of importance-guided selection.

Task complexity. Strategy specificity scales with reasoning chain complexity: MATH-500 (extended multi-step derivation) shows 15–36× importance/random ratios, while GSM8K (shorter arithmetic chains) and GPQA (cross-domain knowledge retrieval) show 2–5×. This is consistent with longer reasoning chains providing more opportunity for discrimination collapse to accumulate differential effects between strategies.

Sparsity level. The imp_vs_rand contrast increases monotonically with sparsity (d_z: 0.16→0.34 on LLaMA), indicating that strategy differentiation is itself dose-dependent—at low sparsity, even importance-based eviction produces minimal compound interaction, compressing the strategy contrast.

5.3 Discrimination Collapse Mechanism

We propose discrimination collapse as the primary mechanism explaining phase-dependent amplification. Under compound pruning, the removal of attention heads degrades the specificity of importance scores used for KV cache eviction. This manifests as declining Gini coefficients and coefficients of variation in importance score distributions within later reasoning phases, where attention patterns become more diffuse.

The discrimination collapse account explains three observations: (1) why compound interaction exceeds the sum of individual effects (pruning degrades the quality of the eviction criterion itself); (2) why later phases are more vulnerable (attention diffusion in consolidation phases reduces score discrimination); (3) why the effect is stronger for importance-based eviction (random eviction is unaffected by score quality degradation).

Table 7: Cross-benchmark synthesis. Phase gradient universality holds across all conditions; strategy specificity is task-modulated.

Metric	MATH-500	GSM8K	GPQA
Models tested	DS+QW	DS+QW	DS
n problems	500	50	50
Imp. gradient (s015→s030)	0.39→8.48	0.24→4.68	0.16→3.15
Dose-response ρ	1.0	1.0	1.0
Bonferroni all sig?	Yes (4/4)	Yes (4/4)	Yes (4/4)
Imp/Rand ratio (DS)	15–36×	2.3–3.8×	3.2–3.7×
Random onset	∼s025	NS all ($n = 50$)	∼s030

Cross-model analysis confirms that Gini/CV decline is consistent across DeepSeek and Qwen, while a secondary entropy redistribution effect is observed only on DeepSeek (Qwen shows direction reversal at s025), suggesting multiple interacting mechanisms with model-dependent relative contributions.

5.4 Limitations

Several limitations qualify our findings: (1) All models are at 7–8B scale; larger models may exhibit different compound interaction dynamics. (2) Strategy specificity magnitude is task-dependent, and the full landscape of task characteristics driving attenuation remains uncharacterized. (3) The LLaMA random phase gradient mechanism (hypothesized GQA effect) lacks direct causal evidence. (4) Phase segmentation via token position quantiles is a proxy; more principled phase detection could reveal finer-grained structure. (5) We do not measure wall-clock throughput implications—our analysis characterizes severity patterns rather than deployment costs.

6 Conclusion

We present the first systematic quantification of phase-dependent compound sparsity severity across models, architectures, and reasoning benchmarks. Our central finding—that compound interaction severity is universally amplified in later reasoning phases ($d_z \approx 1.0$–1.2)—establishes phase-dependence as a fundamental property of compound inference optimization. The dose-response relationship (Spearman $\rho = 1.0$ across all conditions) and sub-exponential growth provide quantitative foundations for predicting compound degradation at untested sparsity levels.

Strategy specificity is most pronounced under demanding mathematical reasoning with extended inference chains (15–36×), attenuating on simpler or cross-domain benchmarks (2–5×) and LLaMA architecture (3:1 d_z ratio). This architecture-dependent modulation, combined with the universality of the phase gradient itself, motivates phase-aware compound sparsity controllers that adapt eviction aggressiveness based on reasoning phase—a direction we leave for future work.

References

Chen, X., et al. SparK: Joint Sparse Compression for Efficient LLM Inference. *NeurIPS*, 2024.

Feng, Z., et al. ThinKV: Phase-Aware KV Cache Compression for Reasoning Models. *arXiv:2510.01290*, 2025.

Li, H., et al. R-KV: Reasoning-Optimized KV Cache Compression. *arXiv:2505.24133*, 2025.

Wang, Y., et al. Physics of KV Cache Compression: Phase Transitions and Representational Rigidity. *arXiv:2603.01426*, 2026.

Zhang, L., et al. HIES: Entropy-Compensated Importance Scoring for KV Cache Eviction. *arXiv:2510.13832*, 2025.

Park, J., et al. Lethe: Layer-Time-Adaptive KV Cache Pruning. *AAAI*, 2026.

Liu, F., et al. Ada-KV: Adaptive KV Cache Compression. *arXiv:2407.11550*, 2024.